Time horizons: A policy strategy may work as intended in the long term, but it can yield unpleasant side effects in the short term.

People respond to incentives: Individuals, officials and politicians, all respond to incentives. The Lucas critique is at work; behaviour changes once a new policy is introduced. This often changes the world when compared with what was understood at the time the policy analysis was undertaken. This is completely unlike physical systems. The behaviour of air molecules does not change when we use them to put planes aloft. But in social science settings, the empirical regularities which were present in the data are modified when policymakers change incentives and thus behaviour.

Non-linearities: Unintended consequences flow from non-linear science. Non-linearities, and the interactions of a large number of moving parts, are notorious for potentially inducing chaos in physical systems. Social systems suffer from the even greater problem that the moving parts are not fixed creatures but optimizing persons. It would be hard enough to analyse non-linear interactions between 1.3 billion people; it is even harder to do this as each one is a sentient being in the quest for life, liberty and happiness.

Consider it possible you may be mistaken

By default, the public policy process is loath to admit failure and change course. Bureaucracies prefer to stay in their comfort zone, they tend to favour intensification of existing programmes as opposed to deeper reform.

Politicians and officials are loath to admit there was a mistake.

The outcomes will be improved if there are formal processes that encourage questioning. Every intervention into society should be subject to ex post review, to ask the question: Did this deliver on its objectives? Does the ex post cost–benefit analysis line up with the ex ante cost–benefit analysis? Did the benefits, as revealed in operation, actually outweigh the costs as revealed in operation?

Public policy is a process of hypothesis testing, of forming a theory about the world, and experimenting with interventions that are thought to help.